



Telkomsel: Supporting customers' digital lifestyles through a super app with AI-powered search

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GOOGLE CLOUD RESULTS

- ✓ Recorded a 20% increase in purchases
- ✓ Search visitor growth 6x, with 88% rise in clicks
- ✓ Revenue attributed to search increased 5x
- ✓ Reduced human or manual updates to the keyword database to zero
- ✓ Reduced customer complaints by 27%

Leading Indonesian mobile service provider Telkomsel needed to unify search across content and services to turbocharge the upgrade of MyTelkomsel

to a ‘super app’ supporting customers’ digital lifestyles. With Google Cloud AI services such as Vertex AI Search and Gemini, Telkomsel is delivering rapid, personalized search that enhances customer experiences and helps drive growth.

As Indonesia’s largest mobile operator, [Telkomsel](#)’s brand and services are deeply embedded into the lives of the country’s citizens. The business invests heavily in digital technologies to innovate and mobile technologies to improve connectivity across the Indonesian archipelago.

Over the past quarter-century, Telkomsel has focused on digital transformation and as this program matured, the operator decided to upgrade its MyTelkomsel mobile app from a mobile account and service management tool for customers to a super-app that acted as a digital lifestyle hub.

This transformation would enable customers to use the app to manage activities across categories that include personal health, travel, payment, entertainment, commerce, and more. These activities included tracking running performance, booking tickets on transport services, checking flight status, shopping, and paying bills. They would also be able to watch videos, listen to music, read articles, and play games directly on the app.

Bringing these super-app features to more than 40 million MyTelkomsel monthly average users would turbocharge Telkomsel’s ambition to transform customer experiences, enhance user satisfaction and drive business growth.

However, to make the project a success, Telkomsel needed to unify search across a wide range of content and services within MyTelkomsel. This project would entail rectifying a range of issues, including overly precise keyword matching that meant relevant search results were omitted, and low product discovery and purchase using search. Also, updating keywords was a high-maintenance process, and users struggled with varied keywords and keyword associations.

These issues made delivering a seamless, personalized customer experience difficult. Customers found it hard to navigate the app and find relevant information, hampered by the need for exact keyword matching within the search function. This often led to frustrating experiences for users if they did not use the precise terminology for packages or services, ultimately resulting in missed opportunities and lost revenue.

Additionally, the Telkomsel backend IT team faced a significant burden, as they were required to constantly update and expand the keyword database manually to improve search accuracy, and could not recommend products and services that matched individual customer preferences effectively.

Telkomsel elected to integrate generative AI into MyTelkomsel search through [Gemini](#) and [Vertex AI](#). “We chose Google Cloud because of its expertise in AI and cloud computing, which aligned perfectly with our digital transformation goals at Telkomsel,” says Ricky Roland, General Manager, Digital Journey and Service Design, at Telkomsel. “Google Cloud is renowned for its robust infrastructure and advanced AI capabilities, particularly through Gemini and Vertex AI Search, which allows businesses to deploy cutting edge machine learning models and enable AI-driven personalization at scale.”

Telkomsel announced the initiative as part of a broader program to infuse Google Cloud AI and data services into its business, customer, and employee-facing applications.

“We saw Google Cloud as a trusted partner to make MyTelkomsel smarter. This was achieved by building an AI-powered search feature that delivered more relevant results faster and tailored to our customer needs. With Gemini and Vertex AI Search, we can recommend products and services that match our customer preferences.”

Ricky Roland

General Manager, Digital Journey and Service Design, Telkomsel



Enhancing the customer experience with AI

“We saw Google Cloud as a trusted partner to make MyTelkomsel smarter,” adds Roland. “This was achieved by building an AI-powered search feature that delivered more relevant results faster and tailored to our customer needs. With Gemini and Vertex AI, we can recommend products and services that match our customer preferences.

“Google Cloud and Gemini provided an integrated environment to manage customized models, allowing us to continuously innovate and enhance the customer experience,” he adds. “With their support, we have integrated AI into the MyTelkomsel app, greatly enhancing the personalized experience and user engagement.”

Working with [Devoteam](#), Telkomsel fully integrated Vertex AI Search and Gemini into the MyTelkomsel app in June 2024. This process entailed migrating data from the organization’s [BigQuery](#) and [Datastore](#) environments into Vertex AI Search to realize the project’s objectives. The mobile operator is working with Google Cloud on a roadmap to introduce more personalized features, including recommendations, and integrate additional data categories, including those covering Indihome fixed-line products, into AI-powered search for MyTelkomsel.

Vertex AI Search and Gemini now support accurate, personalized search across multimodal data, including text, audio, video, images, and more. Navigation is simplified and streamlined, greatly improving user satisfaction. The business also leveraged the tools to launch a virtual assistant that provided instant, relevant answers to customer queries through auto-completion, natural language understanding, and search results tailored to individual usage patterns. This reduced wait times and freed up MyTelkomsel’s human agents to solve more complex problems. Engineers also no longer have to do manual updates to the keyword search database.

Within three months, Telkmosel had recorded an 88% rise in clicks and a 20% increase in purchases across MyTelkomsel. “Users are now relying heavily on search features to find telecommunications products,” says Roland. “When we look at customer segments by length of stay with us, we see these features being particularly

useful for customers that have been with us for one year, which enables us to fight churn. We also see that newer customers who have stayed with us for one to three months or three to six months to date are increasingly using search.

“This indicates that AI-powered search effectively improves the experience for long-term and new users.”

Considerable improvements in search engagement and revenue from search interaction within MyTelkomsel further bear out this outcome. “Our partnership with Google Cloud using Gemini has enhanced our customer experience through AI and machine learning,” says Roland. “The result is very very impressive. It has led to a 355% increase in search engagement and a more than three times revenue boost from search interaction in the MyTelkomsel platform itself.”

Telkomsel views Google Cloud AI tools as critical to its plans to further enhance search within the MyTelkomsel apps. “Later, we will integrate Gemini and Vertex AI Search to deliver faster and more relevant search to customers, based on their preferences and behaviors,” says Roland.

“For example, future versions will not only provide relevant search, but personalized recommendations, whether that be a telecommunications package or channel service such as a channel subscription,” he added. “This is the key to solving current challenges, such as search relevance and lack of structured recommendations. This AI-driven enhancement will ensure MyTelkomsel evolves into a platform that anticipates customer needs, boosting user satisfaction and engagement.

“At the same time, it will strengthen Telkomsel’s leadership in digital transformation across Asia-Pacific, helping us drive growth and innovation in the region.”

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Providing helpful answers with little fuss

Beyond the search experience, Telkomsel goes further with its customer service platform, the Digital Smart Care (DSC) chatbot. A feature-rich avenue in which to address customer needs and handle complaints, it enables agents to deliver quicker and more accurate responses to customers without extensive product training.

Harnessing the power of GenAI through Gemini and its advanced natural language chat capabilities and the search capability of [Vertex AI Search](#), DSC can seamlessly access Telkomsel's Knowledge Management System (KMS), precisely retrieving the right answers and drawing from critical data instantaneously, helping agents with the customers.

The use of DSC reduces wait times and the need for human interaction, effectively cutting complaints down by 27% and saving up to 20% of time and resources otherwise needed for employee training. Agents can focus more on supporting more complex requests and improving the overall service quality, while customers can count on support throughout the day.

Telkomsel's AI investments complement those in Google Cloud infrastructure, enabling mobile communications providers to optimize their 5G and IoT services. "Overall, the partnership with Google Cloud has been key to accelerating innovation and delivering more value to our customers," concludes Roland.

A subsidiary of PT Telekomunikasi Indonesia Tbk and Singapore Telecom Mobile Pte Ltd (Singtel Mobile), **Telkomsel** is Indonesia's largest mobile service provider. Founded in 1995, the business has almost 160 million customers across the archipelago.

Industry: Technology

Location: Indonesia

Products: [Gemini](#), [Vertex AI Search](#), [BigQuery](#), [Datastore](#)

About Google Cloud partner — [Devoteam](#)

[Devoteam](#) provides Google Cloud training, consulting, managed services and services to help companies and teams solve their most complicated business challenges. The business has 2,300+ customers across 20+ countries.

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